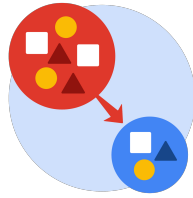


Course Three

From Data to Insight: The Power of Statistics



Instructions

Use this PACE strategy document to record decisions and reflections as you work through this end-of-course project. As a reminder, this document is a resource that you can reference in the future, and a guide to help you consider responses and reflections posed at various points throughout projects.

Course Project Recap

Regardless of which track you have chosen to complete, your goals for this project are:

- Complete the questions in the Course 3 PACE strategy document
- Answer the questions in the Jupyter notebook project file
- Compute descriptive statistics
- Conduct a hypothesis test
- Create an executive summary for external stakeholders

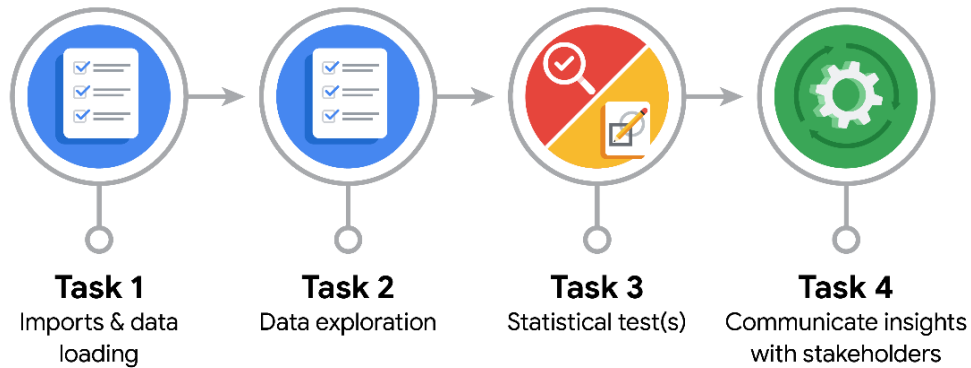
Relevant Interview Questions

Completing this end-of-course project will empower you to respond to the following interview topics:

- How would you explain an A/B test to stakeholders who may not be familiar with analytics?
- If you had access to company performance data, what statistical tests might be useful to help understand performance?
- What considerations would you think about when presenting results to make sure they have an impact or have achieved the desired results?
- What are some effective ways to communicate statistical concepts/methods to a non-technical audience?
- In your own words, explain the factors that go into an experimental design for designs such as A/B tests.

Reference Guide

This project has four tasks; the visual below identifies how the stages of PACE are incorporated across those tasks.



Data Project Questions & Considerations



PACE: Plan Stage

- What is the main purpose of this project?

The purpose of this project is to determine whether the TikTok account status affects the viewer count.

- What is your research question for this project?

Is there a statistically significant difference in the number of viewers for TikTok videos posted by verified and unverified accounts?

- What is the importance of random sampling?

Random sampling helps eliminate biases and ensures that the sample accurately reflects the characteristics of the population by giving every member an equal chance to be selected.

- Give an example of sampling bias that might occur if you didn't use random sampling.

A sampling bias occurs when 80-90% of the subjects belong to a specific category and only 10% to the other. In our example, a sample disproportionately contained more video from verified creators than from unverified ones. This would lead to the conclusion that there is no difference in the viewer counts between verified and unverified accounts, while in reality, there is.



PACE: Analyze & Construct Stages

- In general, why are descriptive statistics useful?

Descriptive statistics help data professionals explore and gain a better understanding of the data (especially large datasets) by showing where the center of the data is, how spread the data is from the mean, and the position of a specific data value in relation to other data values.

- How did computing descriptive statistics help you analyze your data?

Computing descriptive statistics helps me explore and gain an initial understanding of the dataset. By using the pandas `describe()` method, we can see the total number of entries, along with the mean, standard deviation, maximum, and minimum values for numerical data. For categorical data, it provides the number of unique values, the most frequent value (top), and its frequency. In other words, computing these statistics provides a thorough summary of the data that guides further analysis

- In hypothesis testing, what is the difference between the null hypothesis and the alternative hypothesis?

The null hypothesis is a statement that is assumed to be true unless there is convincing evidence to the contrary.

The alternative hypothesis is a statement that contradicts the null hypothesis and is accepted as true only if there is convincing evidence for it.

- How did you formulate your null hypothesis and alternative hypothesis?

Null hypothesis: there is no difference in the number of viewers for TikTok videos posted by verified and unverified accounts. That means any observed difference is due to chance.

Alternative hypothesis: there is a difference in the number of viewers for TikTok videos posted by verified and unverified accounts.



- What conclusion can be drawn from the hypothesis test?

Based on the result of the hypothesis test, there is a statistically significant difference in the average number of viewers for verified and unverified accounts.



PACE: Execute Stage

- What key business or organizational insight(s) emerged from your A/B test?

The key business insights that emerged from the A/B test are that unverified accounts have much higher viewer counts than verified accounts. Based on a two-sample t-test, this difference is statistically significant at a 5% significance level.

- What recommendations do you propose based on your results?

More research and investigation should be conducted to determine the root cause of this behavioral difference. For example, are unverified accounts associated with spam bots that play a role in increasing the viewer counts?